EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 7B

1. Steady and Unsteady States

Consider a system whose state evolves according to $\mathbf{x}[k+1] = \mathbf{M}\mathbf{x}[k]$, where

$$\mathbf{M} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}.$$

In this problem, we will use the eigenvalues and eigenvectors of $\mathbf{M}$ to predict the long-term behavior of the system.

(a) For each eigenvalue below, find a nonzero eigenvector $\mathbf{v}_i$ of $\mathbf{M}$ and give its eigenspace $\text{span}(\mathbf{v}_i)$:

i. $\lambda_1 = 1$
ii. $\lambda_2 = 2$
iii. $\lambda_3 = \frac{1}{2}$

(b) Define $\mathbf{x} = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2 + \gamma\mathbf{v}_3$, a linear combination of the eigenvectors. For each of the cases in the table, determine if

$$\lim_{n\to\infty} \mathbf{M}^n\mathbf{x}$$

converges. If it does, what does it converge to?

α	β	γ	Converges?	$\lim_{n\to\infty}\mathbf{M}^n\mathbf{x}$
0	0	$\neq 0$		
0	$\neq 0$	0		
0	$\neq 0$	$\neq 0$		
$\neq 0$	0	0		
$\neq 0$	0	$\neq 0$		
$\neq 0$	$\neq 0$	0		
$\neq 0$	$\neq 0$	$\neq 0$		

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2. Dive into the Stage

David Copperfield has invented a new illusion! In his new trick, he teleports from one place on a three-dimensional stage to another. The teleporter is represented by the 3×3 matrix $\mathbf{T}$: if David starts at 3D location $\mathbf{s}[t]$, he ends up at

$$\mathbf{T}\,\mathbf{s}[t] = \mathbf{s}[t+1].$$

David says the matrix $\mathbf{T}$ is his secret, but he can tell us its eigenvalue-eigenvector pairs:

$$\lambda_1 = 1, \quad \mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} \qquad \lambda_2 = \frac{1}{3}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} \qquad \lambda_3 = 13, \quad \mathbf{v}_3 = \begin{bmatrix} -3 \\ 5 \\ 0 \end{bmatrix}$$

(a) David starts at $\mathbf{s}[0] = \begin{bmatrix} 8 \\ 2 \\ -5 \end{bmatrix}$. Write $\mathbf{s}[0]$ as a linear combination of the eigenvectors of $\mathbf{T}$.

For the rest of the problem, assume David starts at the same place as in part (a): $\mathbf{s}[0] = \begin{bmatrix} 8 \\ 2 \\ -5 \end{bmatrix}$.

(b) Where will he be at $t = 50$?

(c) David's final trick is that his show will last forever. Will he converge to a single location after infinite time steps? Justify your answer.

(d) Would convergence still occur if the initial position had a small nonzero component along $\mathbf{v}_3$? Explain.

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